Early Crop Stress Detection Using Multispectral Features and Machine Learning (Documentation)

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1. What This Project Is About

The main goal of this project is **crop health monitoring**, but more specifically, I am focusing on **detecting stress early**, before it becomes clearly visible to the human eye. The idea is that if stress is already obvious in the field, then using drones, multispectral data, or machine learning does not really add much value. The real value is in catching subtle signs early.

The dataset used here is a **synthetic multispectral dataset**. It includes vegetation indices, reflectance-based features, and spatial grid coordinates. Together, these try to represent different parts of how a crop behaves, like greenness, chlorophyll response, moisture sensitivity, and how stress might be spread across the field.

The overall flow of this project is simple: first understand the features, then explore how they behave and relate to each other, and finally train and compare machine learning models to see how well they can classify stressed vs healthy crop areas.

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2. Understanding the Dataset and Features

The dataset contains a mix of vegetation-related and supporting features. Some features directly describe plant health, while others give context about the environment or the field layout.

Feature Groups

- **Vegetation Indices:**
  Mean NDVI, Min NDVI, Max NDVI, NDVI Standard Deviation, GNDVI, EVI, SAVI, Red Edge 1, Red Edge 2

- **Supporting / Biophysical Features:**
  Moisture Index, Canopy Density, NIR Reflectance

- **Spatial Features:**
  Grid X, Grid Y

- **Target Variable:**
  Crop Health (Healthy / Stressed)

Each index captures something slightly different. NDVI-based features mainly describe overall greenness and canopy condition. GNDVI and Red Edge features are more sensitive to chlorophyll-related changes, which often show up earlier when a plant starts getting stressed. Moisture and NIR-related features give more context about water and reflectance behavior.

How I Thought About Feature Selection

Instead of using all features in the model, I tried to **think in terms of what the model should actually learn**. Since early detection is the main goal, I separated the features into two ideas:

- Features that are more about **early physiological stress signals** (like GNDVI, NIR reflectance, Red Edge 1, Moisture Index, and NDVI standard deviation)
- Features that describe **overall vegetation condition** (like Mean NDVI, EVI, and canopy-related features)

This way, I could test whether early stress signals alone are enough, or whether the model really needs general health information to perform well.

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3. Data Loading and Cleaning

I started by loading the dataset and doing basic checks to make sure the data is usable.

What I did here:

- Checked for missing values
- Checked for duplicate rows
- Looked at summary statistics to understand feature ranges
- Verified data types for all columns
- Checked the class distribution of the target variable

The target variable was categorical, so I encoded it. This was the only column that needed encoding.

This step was mainly to make sure that nothing strange goes into the models later.

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4. Exploratory Data Analysis (EDA)

4.1 Feature Distributions

I plotted histograms for all features to see how their values are distributed. Since this is a synthetic dataset, almost all features show **flat and uniform distributions**. There are no strong peaks or heavy tails.

This already suggested that no single feature is dominating the target. Instead, the model will probably rely more on **combinations of features rather than individual ones**.

(Insert Histogram Figure Here)

4.2 Correlation Heat Map

Next, I plotted a correlation heat map to see how features relate to each other. A strong pattern appeared in the vegetation indices, especially among the NDVI family, GNDVI, and EVI.

Many of these features are **almost perfectly correlated**, which means they are carrying very similar information. This also raised the issue of **multicollinearity**, especially for linear models like logistic regression.

(Insert Correlation Heat Map Here)

4.3 Feature Subset Analysis

To understand feature behavior better, I created two conceptual subsets and analyzed them using pair plots.

Early Stress Feature Subset

This subset includes:

- GNDVI
- NIR Reflectance
- Red Edge 1
- Moisture Index
- NDVI Standard Deviation

From the pair plots, **GNDVI stood out clearly**. It was the only feature that showed good independent class separation. The other features did not separate the classes well on their own, but they still add context when combined with GNDVI.

This suggested that early stress information in this dataset is mainly encoded along the GNDVI direction, while the other features mostly play a supporting role.

(Insert Pair Plot – Early Stress Subset Here)

Overall Condition Feature Subset

This subset includes:

- Mean NDVI
- EVI
- Canopy and environmental features

Here, Mean NDVI and EVI were the most dominant features for class separation. The remaining features were more dependent and did not show strong independent discrimination.

This matched the idea that these features describe **general crop condition rather than early stress signals**.

(Insert Pair Plot – Overall Condition Subset Here)

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5. Modeling Strategy

I used two different models to look at the problem from two perspectives:

- **Logistic Regression** – as a simple, interpretable, linear baseline
- **Random Forest** – as a nonlinear model that can learn feature interactions

For fairness and stability, I used:

- Stratified train–test split
- Stratified k-fold cross-validation on the training set
- SMOTE inside the training pipeline to handle class imbalance

The main metrics I focused on were:

- Accuracy
- Recall (most important, because false negatives are costly)
- Precision
- F1-score
- ROC–AUC

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6. Logistic Regression Results

Cross-Validation and Test Metrics

Stratified k-fold cross-validation on the training set gave:

- **Mean Recall:** ~83%
- **Standard Deviation:** ~0.02

On the held-out test set, the model achieved:

- **Accuracy:** ~84%

- **Recall:** ~88%
- **Precision:** ~72%
- **F1-score:** ~79%
- **ROC–AUC:** ~0.91

These results showed that even a simple linear model performs quite well for this dataset.

Threshold Tuning

Since missing stressed crops is worse than raising a few false alarms, I tested different decision thresholds.

At thresholds of 0.3, 0.4, and 0.5, I observed the trade-off between recall and precision. I selected **0.4** as a balance that keeps recall high while not dropping precision too much.

Feature Coefficients

I also looked at the feature coefficients to understand how the model is making decisions. NDVI standard deviation and NIR reflectance had positive influence, while GNDVI and some other indices had negative weights.

This helped interpret how variability and reflectance-based features shape the linear decision boundary.

Logistic Regression Stress Map

Using the predicted probabilities and the spatial grid coordinates, I created a **field-level stress heat map**. This map shows which parts of the field have higher predicted stress risk.

(Insert Logistic Regression Stress Map Here)

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7. Random Forest Results

Cross-Validation and Tuning

The initial stratified cross-validation showed moderate performance. I then used grid search to tune key parameters such as:

- Number of trees
- Maximum depth

- Minimum samples per leaf

After tuning, the overall performance improved.

Test Metrics

On the held-out test set, the tuned random forest achieved:

- **Accuracy:** ~86%
- **Recall:** ~89%
- **Precision:** ~75%
- **F1-score:** ~81%
- **ROC–AUC:** ~0.92

This showed a small but consistent improvement over the baseline model.

Feature Importance

Feature importance analysis showed that **GNDVI was the most important feature** in the random forest, while NDVI standard deviation had the least importance.

This was almost the opposite of what I saw in logistic regression, which highlights how nonlinear models rely on features differently.

Random Forest Stress Map

Using the same threshold of 0.4, I generated a stress probability heat map for the random forest model. This made it easier to compare spatial predictions between both models.

(Insert Random Forest Stress Map Here)

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8. Model Comparison

I created a summary table comparing both models using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC–AUC

Overall, both models performed well. The random forest showed slightly better accuracy, precision, and F1-score, while logistic regression had a slightly higher ROC–AUC.

This suggests that nonlinear feature interactions add some value, but the simpler linear model is still very strong and easier to interpret.

(Insert Model Comparison Table Here)

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9. Field-Level Use and Inspection Strategy

The stress heat maps can be used as a guide for field or drone inspection:

- **High stress areas:** Immediate close inspection
- **Medium stress areas:** Follow-up monitoring
- **Low stress areas:** Routine scanning

This helps focus time and resources on the most critical zones instead of scanning the entire field equally.

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10. Limitations and What Could Be Better

This project uses a synthetic dataset, so it does not capture real-world issues like sensor noise, weather variation, or seasonal effects.

Things that could improve this work:

- Using real drone or satellite data
- Adding time-based data to track stress progression
- Including weather and soil conditions
- Trying more models or ensemble methods

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11. Final Thoughts

This project shows how multispectral features and machine learning can be used to detect early crop stress and visualize it across a field. Both linear and nonlinear models performed well, but in slightly different ways.

The stress maps were especially useful because they turn model predictions into something that can actually guide real-world inspection and decision-making.

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12. Figures and Tables

(Screenshots or figures to include:)

- Feature Distribution Histograms
- Correlation Heat Map
- Pair Plots (Early Stress and Overall Condition Subsets)
- Confusion Matrices
- ROC Curves
- Stress Heat Maps (Both